1) CREATE THE DATABASE

CAPSTONE.sql - LAP...R8JLA\Cell C (64))* CAPSTONE QUERIES....R8JLA\Cell C (63))*

```sql
CREATE DATABASE SuperMart_Db;
USE SuperMart_Db;

CREATE TABLE Customers(CustomerId INT PRIMARY KEY,
                        FirstName VARCHAR(100) NOT NULL,
                        LastName VARCHAR(50) NOT NULL,
                        City  VARCHAR(50) NOT NULL,
                        Phone VARCHAR(10),
                        Email VARCHAR(50) NOT NULL);


CREATE TABLE Orders(OrderId INT IDENTITY(101,1)PRIMARY KEY,
                    CustomerId INT FOREIGN KEY REFERENCES Customers,
                    OrderDate VARCHAR(20),
                    StatusCode VARCHAR(10),
                    TotalAmount FLOAT);
```

2) POPULATE DATABASE

```sql
INSERT INTO Customers
VALUES
(1001,'Slindokuhle','Dumakude','Pretoria','071-155-4896','nothandosli2003@gmail.com'),
(1002, 'Amahle','Dlamini','Durban','081-485-3212','mahle58@gmail.com'),
(1003,'Kelebogele','Mosia','Johannesburg','064-257-3256','kelemosia32@outlook.com'),
(1004,'Tshepo', 'Masemola','Johannesburg','082-258-4537','tkmasemola11@ymail.com'),
(1005,'Thuso','Modiadia','Pretoria','074-762-8965','tmodiadia2002@gmail.com'),
(1006,'Mandisa','Buthelezi','Pretoria',NULL,'msbuthelezi53@gmail.com'),
(1007, 'Mandla','Ndlovu','Pietermaritzburg','076-214-5456','mnndlovu1997@ymail.com')

INSERT INTO Orders
VALUES
(1002,'2026-01-25','D',500),
(1003,'2026-02-15','C',1000),
(1004,'2026-02-27','D',250),
(1006,'2026-03-06','D',2500),
(1003,'2026-03-11','D',700),
(1007,'2026-03-20','D',450),
(1002,'2026-04-10','C',650),
(1004,'2026-05-30','D',1100),
(1007,'2026-06-17','D',990),
(1006,'2026-07-25','P',880)
```

3) BASIC DATA RETRIEVAL

SQLQuery3.sql - LA...BR8JLA\Cell C (63))* CAPSTONE.sql - LAP...R8JLA\Cell C (74))*

```sql
USE SuperMart_Db;

SELECT CustomerId, CONCAT(FirstName, LastName)
AS CustomerName, City, COALESCE(Phone,'No Phone Number') AS PhoneNumber
FROM Customers;
```

99 %

Results | Messages

	CustomerId	CustomerName	City	PhoneNumber
1	1001	SlindokuhleDumakude	Pretoria	071-155-4896
2	1002	AmahleDlamini	Durban	081-485-3212
3	1003	KelebogeleMosia	Johannesburg	064-257-3256
4	1004	TshepoMasemola	Johannesburg	082-258-4537
5	1005	ThusoModiadia	Pretoria	074-762-8965
6	1006	MandisaButhelezi	Pretoria	No Phone Number
7	1007	MandlaNdlovu	Pietermaritzburg	076-214-5456

4) FILTERING DATA

```sql
SELECT *
FROM Orders
WHERE OrderDate BETWEEN '2026-01-01' AND'2026-03-31' ;
```

99 %

Results | Messages

	OrderId	CustomerId	OrderDate	StatusCode	TotalAmount
1	103	1002	2026-01-25	D	500
2	104	1003	2026-02-15	C	1000
3	105	1004	2026-02-27	D	250
4	106	1006	2026-03-06	D	2500
5	107	1003	2026-03-11	D	700
6	108	1007	2026-03-20	D	450

```sql
SELECT CONCAT(FirstName,' ',LastName)
AS CustomerName,Email, City
FROM Customers
WHERE City IN ('Johannesburg','Pretoria')
```

99 %

Results | Messages

	CustomerName	Email	City
1	Slindokuhle Dumakude	nothandosli2003@gmail.com	Pretoria
2	Kelebogele Mosia	kelemosia32@outlook.com	Johannesburg
3	Tshepo Masemola	tkmasemola11@ymail.com	Johannesburg
4	Thuso Modiadia	tmodiadia2002@gmail.com	Pretoria
5	Mandisa Buthelezi	msbuthelezi53@gmail.com	Pretoria

5) SQL JOINS

a. INNER JOIN

```sql
SELECT CONCAT(FirstName,LastName) AS CustomerName, OrderId, OrderDate, TotalAmount
FROM Orders o
INNER JOIN Customers c ON o.CustomerId=c.CustomerId
```

99 %

Results | Messages

	CustomerName	OrderId	OrderDate	TotalAmount
1	AmahleDlamini	103	2026-01-25	500
2	KelebogeleMosia	104	2026-02-15	1000
3	TshepoMasemola	105	2026-02-27	250
4	MandisaButhelezi	106	2026-03-06	2500
5	KelebogeleMosia	107	2026-03-11	700
6	MandlaNdlovu	108	2026-03-20	450
7	AmahleDlamini	109	2026-04-10	650
8	TshepoMasemola	110	2026-05-30	1100
9	MandlaNdlovu	111	2026-06-17	990
10	MandisaButhelezi	112	2026-07-25	880

Inner Join returns, only matching records*(meets the joining condition)* from both tables.

Query execute... | LAPTOP-OMBR8JLA (16.0 RTM) | LAPTOP-OMBR8JLA\Cell C... | SuperMart_Db | 00:00:00 | 10 rows

b) LEFT JOIN

```sql
SELECT CONCAT(FirstName,' ',LastName) AS CustomerName, OrderId, OrderDate, TotalAmount
FROM Customers c
LEFT JOIN Orders o ON c.CustomerId= o.CustomerId;
```

99 %

Results | Messages

	CustomerName	OrderId	OrderDate	TotalAmount
1	Slindokuhle Dumakude	NULL	NULL	NULL
2	Amahle Dlamini	103	2026-01-25	500
3	Amahle Dlamini	109	2026-04-10	650
4	Kelebogele Mosia	104	2026-02-15	1000
5	Kelebogele Mosia	107	2026-03-11	700
6	Tshepo Masemola	105	2026-02-27	250
7	Tshepo Masemola	110	2026-05-30	1100
8	Thuso Modiadia	NULL	NULL	NULL
9	Mandisa Buthelezi	106	2026-03-06	2500
10	Mandisa Buthelezi	112	2026-07-25	880
11	Mandla Ndlovu	108	2026-03-20	450
12	Mandla Ndlovu	111	2026-06-17	990

Left Join preserves the entire left table. It looks for matches on the right table, if none are found, it leaves **NULL.**

Query execute... | LAPTOP-OMBR8JLA (16.0 RTM) | LAPTOP-OMBR8JLA\Cell C... | SuperMart_Db | 00:00:00 | 12 rows

c) RIGHT JOIN

```sql
SELECT CONCAT(FirstName,' ',LastName) AS CustomerName, OrderId, OrderDate, TotalAmount
FROM Customers c
RIGHT JOIN Orders o ON c.CustomerId= o.CustomerId;
```

Results | Messages

	CustomerName	OrderId	OrderDate	TotalAmount
1	Amahle Dlamini	103	2026-01-25	500
2	Kelebogele Mosia	104	2026-02-15	1000
3	Tshepo Masemola	105	2026-02-27	250
4	Mandisa Buthelezi	106	2026-03-06	2500
5	Kelebogele Mosia	107	2026-03-11	700
6	Mandla Ndlovu	108	2026-03-20	450
7	Amahle Dlamini	109	2026-04-10	650
8	Tshepo Masemola	110	2026-05-30	1100
9	Mandla Ndlovu	111	2026-06-17	990
10	Mandisa Buthelezi	112	2026-07-25	880

Right Join preserves the entire right table. It looks for matches on the left table, if none are found, it leaves **NULL.**

d) FULL JOIN

```sql
SELECT CONCAT(FirstName,' ',LastName) AS CustomerName, OrderId, OrderDate, TotalAmount
FROM Customers c
FULL JOIN Orders o ON c.CustomerId= o.CustomerId;
```

Results | Messages

CustomerName	OrderId	OrderDate	TotalAmount
Slindokuhle Dumakude	NULL	NULL	NULL
Amahle Dlamini	103	2026-01-25	500
Amahle Dlamini	109	2026-04-10	650
Kelebogele Mosia	104	2026-02-15	1000
Kelebogele Mosia	107	2026-03-11	700
Tshepo Masemola	105	2026-02-27	250
Tshepo Masemola	110	2026-05-30	1100
Thuso Modiadia	NULL	NULL	NULL
Mandisa Buthelezi	106	2026-03-06	2500
Mandisa Buthelezi	112	2026-07-25	880
Mandla Ndlovu	108	2026-03-20	450
Mandla Ndlovu	111	2026-06-17	990

Full Join returns all rows from both tables, combining rows that meet the join condition and this those that don't.

6) SORTING, AGGREGATION, DATE AND STRING FUNCTION

A) TASK 1

```sql
SELECT UPPER(CONCAT(FirstName,' ',LastName)) AS CustomerName,
        City,LEN(FirstName) AS FirstNameLength
FROM Customers
ORDER BY FirstName ASC;
```

99 %

Results | Messages

	CustomerName	City	FirstNameLength
1	AMAHLE DLAMINI	Durban	6
2	KELEBOGELE MOSIA	Johannesburg	10
3	MANDISA BUTHELEZI	Pretoria	7
4	MANDLA NDLOVU	Pietermaritzburg	6
5	SLINDOKUHLE DUMAKUDE	Pretoria	11
6	THUSO MODIADIA	Pretoria	5
7	TSHEPO MASEMOLA	Johannesburg	6

B) TASK 2

```sql
SELECT City,
       COUNT(*) AS TotalCustomers
FROM Customers
GROUP BY City
ORDER BY TotalCustomers DESC;
```

99 %

Results | Messages

	City	TotalCustomers
1	Pretoria	3
2	Johannesburg	2
3	Pietermaritzburg	1
4	Durban	1

Displaying the number of customers in each City *(Group by)*, using COUNT to calculate the number of rows, giving us the number of customers, then grouping by city.

C) TASK 3

```sql
SELECT COUNT(OrderId) AS NumOrders, AVG(TotalAmount) AS AvrgAmount,
       MAX(TotalAmount) AS HighestAmount, MIN(TotalAmount) AS MinAmount
FROM Orders
```

99 %

Results | Messages

	NumOrders	AvrgAmount	HighestAmount	MinAmount
1	10	902	2500	250

D) TASK 4

```sql
SELECT OrderId, RIGHT(OrderDate,LEN(OrderDate)-CHARINDEX('-',OrderDate,CHARINDEX('-',OrderDate)+1)) AS OrderDay,
        SUBSTRING(OrderDate,CHARINDEX('-',OrderDate)+1,2) AS OrderMonth,
        LEFT(OrderDate, CHARINDEX('-',OrderDate)-1) AS OrderYear
FROM Orders
```

Searching for index of "-" the delimiter

	OrderId	OrderDay	OrderMonth	OrderYear
1	103	25	01	2026
2	104	15	02	2026
3	105	27	02	2026
4	106	06	03	2026
5	107	11	03	2026
6	108	20	03	2026
7	109	10	04	2026
8	110	30	05	2026
9	111	17	06	2026
10	112	25	07	2026

Using **Right**, **Left** & **Substring** to extract day, month, and year from a date. Using **Charindex** within right, left & substring, to find index of a delimiter in the date to extract from and to the index of delimiter

7) SECTION A

CAPSTONE.sql - LAP...R8JLA\Cell C (64)) | CAPSTONE QUERIES....R8JLA\Cell C (63))*

```sql
SELECT c.CustomerId, CONCAT(FirstName, LastName) AS CustomerName, City
FROM Customers c
WHERE CustomerId IN (
SELECT CustomerId
FROM Orders
)

SELECT  c.CustomerId, CONCAT(FirstName, LastName) AS CustomerName, City
FROM Customers c
 WHERE EXISTS(
 SELECT OrderId
 FROM Orders o
 WHERE o.CustomerID = c.CustomerID
)
```

Subqueries, CustomerId that's **IN**/**EXISTS** in Orders.
"Customers that bought at least once"

	CustomerId	CustomerName	City
1	1002	AmahleDlamini	Durban
2	1003	KelebogeleMosia	Johannesburg
3	1004	TshepoMasemola	Johannesburg
4	1006	MandisaButhelezi	Pretoria
5	1007	MandlaNdlovu	Pietermaritzburg

	CustomerId	CustomerName	City
1	1002	AmahleDlamini	Durban
2	1003	KelebogeleMosia	Johannesburg
3	1004	TshepoMasemola	Johannesburg
4	1006	MandisaButhelezi	Pretoria
5	1007	MandlaNdlovu	Pietermaritz...

SECTION B

A) VIEW

```sql
CREATE VIEW CustomerOrders AS
SELECT CONCAT (c.FirstName,c.LastName) AS CustomerName, o.OrderDate, o.TotalAmount
FROM Customers c
INNER JOIN Orders o
ON c.CustomerID = o.CustomerID;
```

99 %

Messages

Commands completed successfully.

Completion time: 2026-08-14T19:27:03.7955028+02:00

View called CustomerOrders, It combines the tables using an INNER JOIN and displays the customer's name…This allows us to retrieve this information easily without rewriting the JOIN query each time.

B) COMMON TABLE EXPRESSION

Creates CTE is a temporary result set that exists only for the duration of the query

```sql
WITH CustomerOrderCount AS
(
    SELECT c.CustomerID,
        CONCAT(c.FirstName,' '+c.LastName) AS CustomerName,
        COUNT(o.OrderID) AS NumberOfOrders
    FROM Customers c
    INNER JOIN Orders o
    ON c.CustomerID = o.CustomerID
    GROUP BY c.CustomerID,c.FirstName,c.LastName
)
SELECT
    CustomerName,
    NumberOfOrders
FROM CustomerOrderCount;
```

Count the number of orders

groups the orders by customer, and counts how many orders each customer has

The final SELECT retrieves the customer's name and their total number of orders from the CTE.

99 %

Results | Messages

	CustomerName	NumberOfOrders
1	Amahle Dlamini	2
2	Kelebogele Mosia	2
3	Tshepo Masemola	2
4	Mandisa Buthelezi	2
5	Mandla Ndlovu	2

SECTION C

```sql
CREATE PROCEDURE GetCustomerOrders
    @CustomerID INT
AS
BEGIN
    SELECT OrderID, OrderDate, StatusCode, TotalAmount
    FROM Orders
    WHERE CustomerID = @CustomerID;
END;

EXEC GetCustomerOrders @CustomerID = 1003;
```

Creates procedure-a saved SQL program that can be executed whenever needed. that accepts a CustomerID as an input parameter

Searching for orders made by customer ID: **1003**

99 %

Results | Messages

	OrderID	OrderDate	StatusCode	TotalAmount
1	104	2026-02-15	C	1000
2	107	2026-03-11	D	700